

JMO – 2010

Max Mark – 100

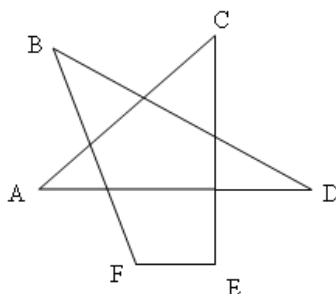
Time – 3 Hrs

Instructions:

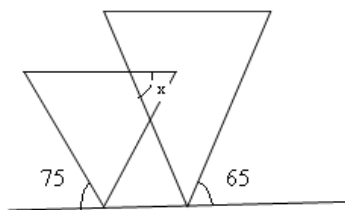
- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
- Q1 to Q 6 compulsory. Answer any one from Q 7 and Q 8.

1. .

- (a) In a contests to guess the number of balls in a box, A guessed 25, B guessed 31, C guessed 29, D guessed 27 and E guessed 23. Two guesses were wrong by 2 and two guesses were wrong by 4, the other guess was correct. Determine the number of balls in the basket. [4]
- (b) The numbers 72, 8, 24, 10, 5, 45, 36, 15 are grouped in pairs so that product of each pair is same. Which number is paired with 10? [4]
- (c) In the figure below determine the sum of angles: $\angle A, \angle B, \angle C, \angle D, \angle E, \angle F$. [4]



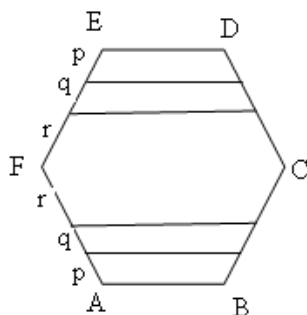
- (d) In the figure the two triangles drawn are equilateral triangles. What is the value of x ? [4]



- (e) If two of the angles of polygon are 90° each and remaining angles are 120° each then find the number of sides of the polygon. [4]
- (f) A watch always shows right time but has only hour hand and minute hand is not there. At the precise moment this hour hand points directly to 16 minute mark, what is the exact time? [4]

2.

- (a) A list of 5 positive integers has all of the following properties: [6]
- (i) The only integers in the list that occurs more than once is 8.
 - (ii) The average of these integers is 10.
 - (iii) When the 5 integers are arranged in non decreasing order the middle one is 9.
- What is the largest possible integer that appear in the list?
- (b) In triangle ABC, $AC = BC$, the bisector of angle $\angle BAC$ meets BC at E, and AE is perpendicular to BC. Determine the measures of all angles of triangle. [6]
- (c) In the adjoining figure the regular hexagon has been divided into four trapezia and one hexagon. If each of five sections has the same perimeter, what is the ratio of the lengths p, q and r? [6]



- (d) A 24 hour digital clock is kept on a glass table top. At a certain time the display in the clock and the reflection of the display by the glass is observed to be the same. How many times in 24 hour period do the display and its reflection gives same time? [6]
- (e) On Monday, A wants to work by a car at an average speed of 70km/hour and arrived 1 minute late. On Tuesday, he left at the same time and took the same route. This time he drove at 75km/hour and arrived 1 minute early. How long is his route to work? [6]
- (f) A shopkeeper cheats 10% both the seller while buying and the purchaser while selling by using a defective balance. What is the % of profit? [6]
3. For non negative distinct integers j, k, l, m are such that $5^j + 6^k + 7^l + 11^m = 2006$. Determine $j + k + l + m = ?$ [8]
4. In the year there were exactly four Tuesday and exactly four Friday in October. On what day of the week did October 31st fall that year. [8]
5. A palindrome is a positive integer that is same when read forward and backwards. For example 21012 and 77277 are palindromes. [8]
- (i) Determine the number of palindrome less than 100.
- (ii) Determine the number of palindrome with 5 digits.
6. If m is a positive integer. $m!$ is used to represent product of integers from 1 to m . Some positive integers can be written in the form $n = a(1!) + b(2!) + c(3!) + d(4!)$, where $0 \leq a \leq 1, 0 \leq b \leq 2, 0 \leq c \leq 3, 0 \leq d \leq 4$. [8]
- (i) Determine the largest positive integers N that can be written in this form
- (ii) Write 105 in this form.
- (iii) Prove that all integers n , $0 \leq n \leq N$, can be written in this form.
7. In a game Saurav and Sunil take turns calling out whole numbers. The first number called must be a whole number between and including 1 and 9. Each number called after the first must be a whole number which is 1 to 10 greater than the previous number called. [8]
- (i) The first time the game is played, the person who calls the number 15 is the winner. Explain why Saurav has a winning strategy if he goes first and calls 4.
- (ii) The second time the game is played, the person who calls the number 50 is the winner. If Saurav goes first, how does he guarantee that he will win?
8. One of persons X, y, Z is teacher and other two are doctor and police officer. Z is older than police officer. X and doctor are not same age and doctor is younger than Y. Decide who is teacher among them? [8]

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